Dongryul M. Kim

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RESEARCH INTEREST

Homogeneous dynamics & Hyperbolic geometry
- Discrete subgroups of Lie groups – Kleinian groups and Anosov representations
- Deformations and Rigidity
- Dynamical systems and Ergodic theory

Geometric topology & Low-dimensional topology
- Hyperbolic 3-manifolds
- Mapping class groups, their subgroups, and dynamics on Teichmüller spaces and Complexes of curves
- Geometric group theory
- Random walks on groups and their stationary measures

EMPLOYMENT

Sep. 2027 – Aug. 2030 **L.E. Dickson Instructor**
The University of Chicago, Chicago, IL, USA

Aug. 2026 – Jul. 2027 **Member**
Institute for Advanced Study, Princeton, NJ, USA

Jan. 2026 – May. 2026 **Gamelin Endowed Postdoctoral Fellow**
Simons Laufer Mathematical Sciences Institute (formerly MSRI), Berkeley, CA, USA
- Program: Geometry and Dynamics for Discrete Subgroups of Higher Rank Lie Groups

EDUCATION

Sep. 2021 – Dec. 2025 **Yale University**, New Haven, CT, USA
Ph.D. in Mathematics, Advisor: Prof. Hee Oh
- Dissertation: Rigidity of Kleinian Groups via Dynamics on Higher Rank Spaces

Mar. 2015 – Feb. 2021 **KAIST**, Daejeon, South Korea
B.S. in Mathematical Sciences (Advanced Major), Summa Cum Laude
- Left for mandatory military service (Mar. 2017 – Dec. 2018)

PUBLICATIONS

32. Higher-rank local mixing for cusped positive representations (with Hee Oh, Wenyu Pan), Preprint (arXiv:2610.12254), Submitted.

31. Extremal entropy for products of Fuchsian representations (with Richard Canary), Preprint (arXiv:2609.10325), Submitted.

30. Strict concavity of the growth indicator function for relatively Anosov groups (with Hee Oh, Andrew Zimmer), Preprint (arXiv:2607.13760), Submitted.

29. Global shadow estimates and Hausdorff measure criteria in higher rank (with Hee Oh), Preprint (arXiv:2606.19779), Submitted.

28. Non-arithmeticity of length spectra of subgroups of mapping class groups (with Inhyeok Choi),
Preprint (arXiv:2605.13064), Merged into the appendix of the published version of paper 22 below in Inventiones Mathematicae.

27. Measurable boundary maps and Patterson–Sullivan measures for non-Borel Anosov groups on the Furstenberg boundary (with Andrew Zimmer),
Preprint (arXiv:2603.28694), Submitted.

26. Vector-valued horofunction boundaries and Patterson–Sullivan measures (with Andrew Zimmer),
Preprint (arXiv:2603.28693), Submitted.

25. Classification of horospherical invariant measures in higher rank: The Full Story (with Inhyeok Choi),
Preprint (arXiv:2601.22668), Submitted.

24. Determining subgroups via stationary measures (with Andrew Zimmer),
International Mathematical Research Notices, (2026), no. 12, rnag131.

23. Classification of horospherical invariant measures in higher rank: Teaser (with Inhyeok Choi),
Preprint (arXiv:2510.23365), Teaser of the paper 25 above and not for publication.

22. Invariant Radon measures on the space of measured laminations for subgroups of mapping class group (with Inhyeok Choi),
To appear in **Inventiones Mathematicae**.

21. A note on the singularity conjecture for infinite covolume discrete subgroups (with Andrew Zimmer),
Preprint (arXiv:2508.05756), Submitted.

20. Rigidity for Patterson–Sullivan systems with applications to random walks and entropy rigidity (with Andrew Zimmer),
Preprint (arXiv:2505.16556), Submitted.

19. Horocycles in hyperbolic 3-manifolds with round Sierpiński limit sets (with Minju Lee),
To appear in **Groups, Geometry, and Dynamics**.

18. Conformal measure rigidity and ergodicity of horospherical foliations,
Preprint (arXiv:2404.13727), Submitted.

17. Relatively Anosov groups: finiteness, measure of maximal entropy and reparameterization (with Hee Oh),
Journal für die reine und angewandte Mathematik (Crelle's Journal), Vol. 826 (2025), 91–142.

16. Ahlfors regularity of Patterson-Sullivan measures of Anosov groups and applications (with Subhadip Dey, Hee Oh),
Compositio Mathematica, Vol. 162 (2026), no. 4, 743–791.

15. Ergodic dichotomy for subspace flows in higher rank (with Hee Oh, Yahui Wang),
Communications of the American Mathematical Society, Vol. 5 (2025), 1–47.

14. Properly discontinuous actions, growth indicators and conformal measures for transverse subgroups (with Hee Oh, Yahui Wang),
Mathematische Annalen, Vol. 393 (2025), 2391–2450.

13. Non-concentration property of Patterson-Sullivan measures for Anosov subgroups (with Hee Oh),
Ergodic Theory and Dynamical Systems, Vol. 45 (2025), no. 3, 884–893.

12. Conformal measure rigidity for representations via self-joinings (with Hee Oh),
Advances in Mathematics, Vol. 458 (2024), Paper No. 109992.

11. Rigidity of Kleinian groups via self-joinings: measure theoretic criterion (with Hee Oh),
Journal of Topology, Vol. 18 (2025), no. 3, Paper No. e70035.

10. Rigidity of Kleinian groups via self-joinings (with Hee Oh),
Inventiones Mathematicae, Vol. 234 (2023), no. 3, 937–948.

9. Reducible normal generators for mapping class groups are abundant (with Hyungryul Baik, Chenxi Wu),
Journal of Topology and Analysis, Vol. 17 (2025), no. 5, 1313–1323.

8. Hausdorff dimension of directional limit sets for self-joinings of hyperbolic manifolds
(with Yair N. Minsky, Hee Oh),
Journal of Modern Dynamics, Vol. 19 (2023), 433–453.

7. Tent property of the growth indicator functions and applications (with Yair N. Minsky, Hee Oh),
Geometriae Dedicata (Labourie's volume) Vol. 218 (2024), no. 1, Paper No. 14, 18 pp.

6. Minimal asymptotic translation lengths on curve complexes and homology of mapping tori
(with Hyungryul Baik, Chenxi Wu),
Michigan Mathematical Journal, Vol. 75 (2025), no. 5, 927–942.

5. Linear growth of translation lengths of random isometries on Gromov hyperbolic spaces and Teichmüller spaces (with Hyungryul Baik, Inhyeok Choi),
Journal of the Institute of Mathematics of Jussieu, Vol. 23 (2024), no. 4, 1751–1795.

4. Simple length spectra as moduli for hyperbolic surfaces and Rigidity of length identities
(with Hyungryul Baik, Inhyeok Choi),
To appear in **Annales de l'Institut Fourier**.

3. On the asymptotic translation lengths on the sphere complexes and the generalized fibered cone
(with Hyungryul Baik, Chenxi Wu),
Transactions of the American Mathematical Society, Vol. 378 (2025), no. 7, 4739–4764.

2. On the surjectivity of the Symplectic representation of the mapping class group
(with Hyungryul Baik, Inhyeok Choi),
Topology and its Applications, Vol. 322 (2022), Paper No. 108334.

1. Topological entropy of pseudo-Anosov maps from a typical Thurston's construction
(with Hyungryul Baik, Inhyeok Choi),
International Mathematical Research Notices, (2022), no. 24, 19862–19904.

Expository writings:

3. Normal generators for mapping class groups
(with Hyungryul Baik)
To appear in *In the Tradition of Thurston: Geometry and Topology*, Vol. 4.

2. Rigidity theorems for hyperbolic 3-manifolds: from finite volume to infinite volume (in Korean)
Horizon, a scientific magazine published by KIAS

1. Kleinian Groups from the Sphere at Infinity and Their Self-Joinings,
Handbook of Visual, Experimental and Computational Mathematics

Talks and Minicourses

Invited minicourses:

1. Thematic semester *Groups, representations and random geometries*, Erdős Center at Renyi Institute Nov. 08–12, 2027 Title: TBD

Invited Talks:

44. Workshop for the thematic semester: Groups, representations and random geometries, Erdős Center at Renyi Institute, Nov. 15–19, 2027
43. Young Mathematician Lecture Series, National University of Singapore, Apr. 21, 2027
42. Joint Mathematics Meetings, AMS Special Session on Discrete Subgroups of Lie groups, Jan. 12–15, 2027
41. George Mason Geometry MMA: Metrics, Measures, and Algorithms Seminar, Nov. 20, 2026
40. University of Texas–Austin Groups and Dynamics Seminar, Nov. 11, 2026
39. PATCH (Philadelphia Area Topology, Contact/Hyperbolic) Seminar, Nov. 06, 2026
38. CUNY Geometry and Topology Seminar, Oct. 27, 2026
37. CUNY Hyperbolic Geometry Seminar, Oct. 27, 2026
36. Midwest Dynamical Systems Conference, Oct. 16–18, 2026
35. AMS Sectional Meeting, Special session on Ergodic Geometry, Oct. 03–04, 2026
34. IAS Short Talks by Postdoctoral Members, Sep. 24, 2026
33. KIAS Virtual Seminar on Geometry and Topology, Jul. 07, 2026 (in person but broadcast)
32. Seoul National University Workshop on Homogeneous dynamics and Number Theory, Jun. 25–26, 2026
31. International Conference for the 80th Anniversary of the Korean Mathematical Society, Jun. 22–25, 2026
30. Rice Dynamics/Geometry Talk, Apr. 08, 2026
29. SLMath Re-GRAD (Results in Group Actions and Dynamics) Seminar, Mar. 19, 2026
28. SLMath Superrigidity Seminar, Mar. 11, 2026
27. SLMath Geometry and Dynamics for Discrete Subgroups of Higher Rank Lie Groups Seminar, Feb. 23, 2026
26. KIAS Virtual Seminar on Geometry and Topology, Dec. 22, 2025 (in person but broadcast)
25. University of Michigan–Ann Arbor Topology Seminar, Dec. 05, 2025
24. University of Michigan–Ann Arbor RTG Geometry, Topology and Dynamics Seminar, Dec. 03, 2025
23. University of Wisconsin–Madison Group Actions and Dynamics Seminar, Oct. 20, 2025
22. Penn State Anatole Katok Center for Dynamical Systems and Geometry Seminar, Sep. 29, 2025
21. Yale Geometry and Topology Seminar, Sep. 09, 2025
20. Yale Undergraduate Math Society Seminar, Sep. 10, 2025
19. KIAS One-day Festival of Group actions, Geometry and Dynamics, Aug. 06, 2025

18. Yale Group Actions and Dynamics Seminar, Apr. 28, 2025
17. Cornell Dynamics Seminar, Apr. 24, 2025
16. Yale Visiting day for Prospective Ph.D. students, Mar. 08, 2025
15. Seoul National University Dynamics seminar, Feb. 19, 2025
14. GT GAPS (Geometric Topology Grad and Postdoc Seminar), Dec. 06, 2025
13. University of Illinois Chicago Ergodic Theory and Dynamical Systems Seminar, Oct. 29, 2024
12. University of Chicago Dynamics Seminar, Oct. 28, 2024
11. University of Wisconsin–Madison Group Actions and Dynamics Seminar, Oct. 21, 2024
10. Meeting with Young Geometric Topologists at KAIST, Jun. 25, 2024
9. Yale Group Actions, Geometry and Dynamics Seminar, Mar. 04, 2024
8. Yale Visiting day for Prospective Ph.D. students, Mar. 10, 2023
7. Yale Group Actions, Geometry and Dynamics Seminar, Oct. 19, 2022
6. Seoul National University Dynamics seminar, Aug. 10, 2022
5. Seoul National University Dynamics seminar, Aug. 09, 2022
4. Yale Visiting day for Prospective Ph.D. students, Mar. 11, 2022
3. CUNY Geometry and Topology Seminar, Nov. 02, 2021
2. Yale Geometry and Topology Seminar, Sep. 21, 2021
1. KIAS Virtual Seminar on Geometry and Topology, Apr. 22, 2020

Contributed Talks:

12. Lightning talk at Geometry and Topology Seminar at Brown and Yale (GATSBY), Nov. 08, 2025
11. Lightning talk at Midwest Summer School in Geometry, Topology, and Dynamics at Madison, Jun. 02, 2025
10. Lightning talk at Geometry and Topology Seminar at Brown and Yale (GATSBY), Apr. 26, 2025
9. Lightning talk at Geometry and Topology Seminar at Brown and Yale (GATSBY), Nov. 16, 2024
8. Moving to higher rank: from hyperbolic to Anosov, A conference in honor of Dick Canary, Jul. 16, 2024
7. Lightning talk at Geometry and Topology Seminar at Brown and Yale (GATSBY), Apr. 20, 2024
6. Workshop on Dynamical systems and related topics at University of Maryland, College Park: in a celebration of Giovanni Forni's 60th birthday, Apr. 07, 2024
5. Graduate Student Seminar at Yale, Jan. 27, 2023
4. Graduate Student Seminar at Yale, Sep. 23, 2022
3. Graduate Student Seminar at Yale, Feb. 11, 2022
2. Graduate Student Seminar at Yale, Sep. 17, 2021
1. Lightning talk at Geometric Group Theory in East Asia, Oct. 30, 2020

Mentoring

General Mentoring

SLMath:

2026 Spring	Mentoring a Ph.D. Student

Directed Reading Program (led reading groups, suggested exercises, assigned projects and presentations)

Yale University:

2025 Fall	Mapping class groups and Graphs of curves
2024 Fall	Mostow's rigidity theorem

Teaching

Yale University:

Primary Instructor

2025 Spring	MATH 115 Calculus of Functions of One Variable II
2024 Spring	MATH 115 Calculus of Functions of One Variable II

Teaching Fellow

2025 Fall	MATH 5260 Introduction to Differentiable Manifolds
2024 Fall	MATH 526 Introduction to Differentiable Manifolds
2023 Fall	MATH 302 Vector Analysis and Integration on Manifolds
2023 Spring	MATH 435 Differential Geometry
2023 Spring	MATH 244 Discrete Mathematics
2022 Fall	MATH 302 Vector Analysis and Integration on Manifolds
2022 Fall	MATH 120 Calculus of Functions of Several Variables

KAIST:

Tutor for freshmen

2020 Fall	MAS102 Calculus II
2020 Spring	MAS101 Calculus I
2019 Fall	MAS102 Calculus II

Awards and Grants

- June E Huh Visiting Fellowship, June E Huh Center for Mathematical Challenges at KIAS, 2026 – 2029
- AMS International Congress of Mathematicians (ICM) 2026 Travel Grant (NSF, DMS–2529637)
- Gamelin Endowed Postdoctoral Fellowship, Simons Laufer Mathematical Sciences Institute, 2026
- Gibbs medal (Department Dissertation Prize), Yale University, 2025
- Overseas PhD Scholarship, Korea Foundation for Advanced Studies, 2021 – 2025
- Undergraduate Scholarship, Korea Foundation for Advanced Studies, 2019–2021
- Korea Presidential Science Scholarship, Korea Student Aid Foundation, 2015 – 2021
- KAIST Presidential Fellowship (KPF), KAIST, 2015 – 2021

Organizational activities

Jan. 2026 – May. 2026 Professional Development Seminar at SLMath
- Organizer

Sep. 2022 – Dec. 2022 Yale Graduate Students Seminars
- Organizer

Jun. 2022 – May. 2023 Yale Korean Graduate Student Association (KGSA)
- President

Aug. 2021 – Jun. 2022 Yale Korean Graduate Student Association (KGSA)
- Treasurer

Feb. 2016 – Jun. 2021 KAIST Undergraduate Math Colloquium
- Organizer

Jan. 2016 – Jan. 2017 KAIST Presidential Fellowship Student Society
- Student President

Services

Quick opinion for GAFA

Referee for Compositio, G&T, TAMS

Sep. 2024 – Dec. 2025 Climate Committee, Dept. of Mathematics at Yale University

Mar. 2017 – Dec. 2018 Republic of Korea Army
- Mandatory military service, Sentinel & Programmer

Mar. 2016 – Dec. 2016 Notices of the KAIST College of Natural Sciences
- Editor, Student Reporter